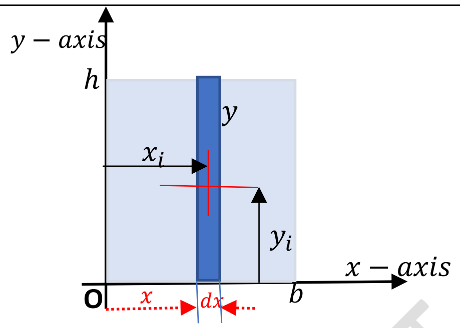
## Example 1

Locate the centroid of the rectangle of dimensions $b$ and $h$ with respect to the co-ordinate axes.




Consider an infinitesimal element of thickness $dx$ and height $y$ at a distance $x$ from the origin. Let the centre of this rectangular element be at $(x_i, y_i)$

$$x_i \cong x \quad \lll\ggg \quad y_i = \frac{y}{2} = \frac{h}{2} \quad \lll\ggg \quad dA = y\,dx = h\,dx$$

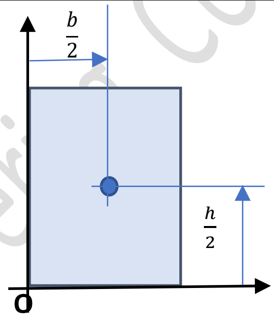
The centroid is given by

$$x_c = \frac{\int x_i\,dA}{\int dA} = \frac{\int_0^b x\,hdx}{\int_0^b hdx} = \frac{h\int_0^b x\,dx}{\int_0^b hdx} = \frac{h\left[\frac{x^2}{2}\right]_0^b}{h[x]_0^b} = \frac{\frac{hb^2}{2}}{bh} = \frac{b}{2}$$

(The integration is with respect to $x$ which varies from $0$ to $b$)

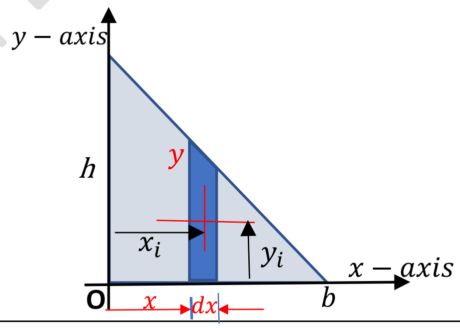
$$y_c = \frac{\int y_i\,dA}{\int dA} = \frac{\frac{h}{2}\int_0^b h\,dx}{\int_0^b hdx} = \frac{\frac{h^2}{2}[x]_0^b}{h[x]_0^b} = \frac{\frac{bh^2}{2}}{bh} = \frac{h}{2}$$

(The integration is with respect to $x$ which varies from $0$ to $b$)




## Example 2

Locate the centroid of the right-angled triangle of dimensions $b$ and $h$ with respect to the co-ordinate axes.




Consider an infinitesimal element of thickness $dx$ and height $y$ at a distance $x$ from the origin. Let the centre of this rectangular element be at $(x_i, y_i)$

$$x_i \cong x \quad \lll\ggg \quad y_i = \frac{y}{2} \quad \lll\ggg \quad dA = y\,dx$$

From the similar triangles,

$$\frac{y}{h} = \frac{b-x}{b} \quad \ggg\ggg \quad y = \frac{h(b-x)}{b}$$

$$\therefore y_i = \frac{h(b-x)}{2b} \quad \lll\ggg \quad dA = \frac{h(b-x)\,dx}{b}$$

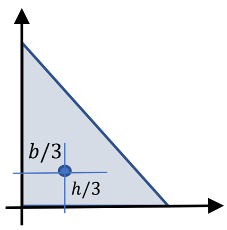
The centroid is given by

$$x_c = \frac{\int x_i\,dA}{\int dA} = \frac{\int_0^b x\,\frac{h(b-x)\,dx}{b}}{\int_0^b \frac{h(b-x)\,dx}{b}} = \frac{\frac{h}{b}\left[\frac{bx^2}{2} - \frac{x^3}{3}\right]_0^b}{\frac{h}{b}\left[bx - \frac{x^2}{2}\right]_0^b} = \frac{\frac{h}{b}\frac{b^3}{6}}{\frac{h}{b}\frac{b^2}{2}} = \frac{\frac{b^2h}{6}}{\frac{bh}{2}} = \frac{b}{3}$$

(The integration is with respect to $x$ which varies from $0$ to $b$)

$$y_c = \frac{\int y_i\,dA}{\int dA} = \frac{\int_0^b \frac{h(b-x)}{2b}\,\frac{h(b-x)\,dx}{b}}{\int_0^b \frac{h(b-x)\,dx}{b}} = \frac{\frac{h^2}{2b^2}\left[b^2x - \frac{2bx^2}{2} + \frac{x^3}{3}\right]_0^b}{\frac{bh}{2}} = \frac{h}{3}$$

(The integration is with respect to $x$ which varies from $0$ to $b$)

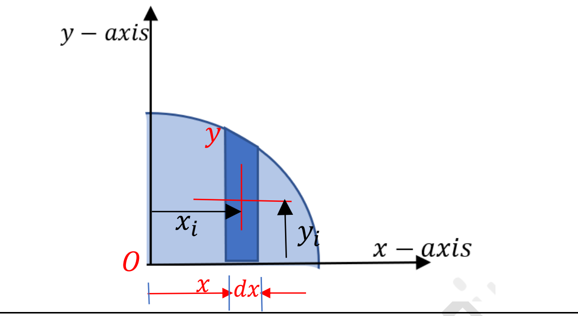


**Example 3**

Locate the centroid of the quarter circular area of radius $r$ with respect to the co-ordinate axes shown.

The equation of the circle whose centre is at the origin and radius $r$ is

$x^2 + y^2 = r^2$ ⋙ $y = \sqrt{r^2 - x^2}$




Consider an infinitesimal element of thickness $dx$ and height $y$ at a distance $x$ from the origin. Let the centre of this rectangular element be at $(x_i, y_i)$

$$x_i \cong x \quad \lll\ggg \quad y_i = \frac{y}{2} = \frac{\sqrt{r^2-x^2}}{2} \quad \lll\ggg \quad dA = y\,dx = \sqrt{r^2 - x^2}\,dx$$

The centroid is given by

$$x_c = \frac{\int x_i\,dA}{\int dA} = \frac{\int_0^r x\sqrt{r^2-x^2}\,dx}{\int_0^r \sqrt{r^2-x^2}\,dx} = \frac{-\int_r^0 t^2\,dt}{\int_0^{\frac{\pi}{2}} \sqrt{r^2 - r^2\sin^2\theta}\,(r\cos\theta\ d\theta)}$$

$$= \frac{\int_0^r t^2\,dt}{r^2\int_0^{\frac{\pi}{2}} \cos^2\theta\,d\theta} = \frac{\left[\frac{t^3}{3}\right]_0^r}{r^2\int_0^{\frac{\pi}{2}} \frac{1-\cos 2\theta}{2}\mathrm{d}\theta} = \frac{\frac{r^3}{3}}{\frac{r^2}{2}\left[\theta - \frac{\sin 2\theta}{2}\right]_0^{\frac{\pi}{2}}}$$

$$= \frac{\frac{r^3}{3}}{\frac{r^2}{2}\frac{\pi}{2}} = \frac{4r}{3\pi}$$

numerator

$r^2 - x^2 = t^2 \gg -2xdx = 2tdt$

$x\,dx = -tdt$

When $x = 0 \to t = r$

When $x = r \to t = 0$

denominator

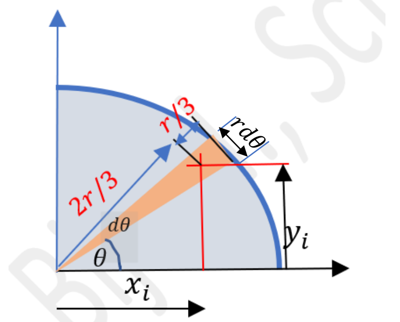
$x = r\sin\theta \gg dx = r\cos\theta\ d\theta$

When $x = 0 \to \theta = 0$

When $x = r \to \theta = \frac{\pi}{2}$

$$y_c = \frac{\int y_i\,dA}{\int dA} = \frac{\int_0^r \frac{\sqrt{r^2-x^2}}{2}\sqrt{r^2-x^2}\,dx}{\int_0^r \sqrt{r^2-x^2}\,dx} = \frac{\frac{1}{2}\int_0^r (r^2-x^2)\,dx}{\int_0^r \sqrt{r^2-x^2}\,dx} = \frac{\frac{1}{2}\left[r^2x - \frac{x^3}{3}\right]_0^r}{\frac{\pi r^2}{4}} = \frac{\frac{1}{2}\frac{2r^3}{3}}{\frac{\pi r^2}{4}} = \frac{4r}{3\pi}$$

We can also solve the above problem by considering a triangular strip of included angle $d\theta$ at angular orientation $\theta$ as shown. The centroid of the strip is at $\frac{r}{3}$ from the base.



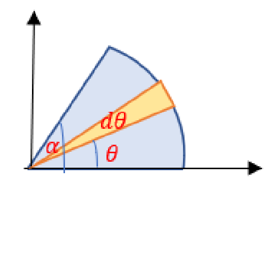

$$x_i = \frac{2}{3}r\cos\theta \quad \lll\ggg \quad y_i = \frac{2}{3}r\sin\theta \quad \lll\ggg \quad dA = \frac{1}{2}r \times rd\theta = \frac{1}{2}r^2 d\theta$$

$$x_c = \frac{\int x_i\,dA}{\int dA} = \frac{\int_0^{\frac{\pi}{2}} \frac{2}{3}r\cos\theta\ \frac{1}{2}r^2 d\theta}{\int_0^{\frac{\pi}{2}} \frac{1}{2}r^2 d\theta} = \frac{\frac{r^3}{3}\int_0^{\frac{\pi}{2}}\cos\theta\ d\theta}{\frac{r^2}{2}\int_0^{\frac{\pi}{2}} d\theta} = \frac{\frac{r^3}{3}[\sin\theta]_0^{\frac{\pi}{2}}}{\frac{r^2}{2}[\theta]_0^{\frac{\pi}{2}}} = \frac{4r}{3\pi}$$

(The integration is with respect to $\theta$ which varies from 0 to $\frac{\pi}{2}$)

$$y_c = \frac{\int y_i\,dA}{\int dA} = \frac{\int_0^{\frac{\pi}{2}} \frac{2}{3}r\sin\theta\ \frac{1}{2}r^2 d\theta}{\int_0^{\frac{\pi}{2}} \frac{1}{2}r^2 d\theta} = \frac{\frac{r^3}{3}\int_0^{\frac{\pi}{2}}\sin\theta\ d\theta}{\frac{r^2}{2}\int_0^{\frac{\pi}{2}} d\theta} = \frac{\frac{r^3}{3}[-\cos\theta]_0^{\frac{\pi}{2}}}{\frac{r^2}{2}[\theta]_0^{\frac{\pi}{2}}} = \frac{4r}{3\pi}$$

A circular sector of included angle $\alpha$



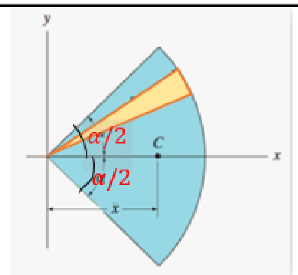

$$x_c = \frac{\int x_i\,dA}{\int dA} = \frac{\int_0^{\alpha} \frac{2}{3}r\cos\theta\ \frac{1}{2}r^2 d\theta}{\int_0^{\alpha} \frac{1}{2}r^2 d\theta} = \frac{\frac{r^3}{3}\int_0^{\alpha}\cos\theta\ d\theta}{\frac{r^2}{2}\int_0^{\alpha} d\theta} = \frac{\frac{r^3}{3}[\sin\theta]_0^{\alpha}}{\frac{r^2}{2}[\theta]_0^{\alpha}} = \frac{2r}{3\alpha}\sin\alpha$$

$$y_c = \frac{\int y_i\,dA}{\int dA} = \frac{\int_0^{\alpha} \frac{2}{3}r\sin\theta\ \frac{1}{2}r^2 d\theta}{\int_0^{\alpha} \frac{1}{2}r^2 d\theta} = \frac{\frac{r^3}{3}\int_0^{\alpha}\sin\theta\ d\theta}{\frac{r^2}{2}\int_0^{\alpha} d\theta} = \frac{\frac{r^3}{3}[-\cos\theta]_0^{\alpha}}{\frac{r^2}{2}[\theta]_0^{\alpha}} = \frac{2r}{3\alpha}(1-\cos\alpha)$$

A circular sector of included angle $\alpha$

$$x_c=\frac{\int x_i\,dA}{\int dA}=\frac{\int_{-\alpha/2}^{\alpha/2}\frac{2}{3}r\cos\theta\ \frac{1}{2}r^2d\theta}{\int_{-\alpha/2}^{\alpha/2}\ \frac{1}{2}r^2d\theta}=\frac{\frac{r^3}{3}\int_{-\alpha/2}^{\alpha/2}\cos\theta\ \ d\theta}{\frac{r^2}{2}\int_{-\alpha/2}^{\alpha/2}\ \ d\theta}=\frac{\frac{r^3}{3}[\sin\theta]_{-\alpha/2}^{\alpha/2}}{\frac{r^2}{2}[\theta]_{-\alpha/2}^{\alpha/2}}=\frac{4r}{3\alpha}\sin\frac{\alpha}{2}$$

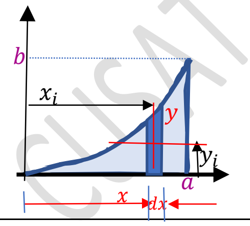
$$y_c=\frac{\int y_i\,dA}{\int dA}=\frac{\int_{-\alpha/2}^{\alpha/2}\frac{2}{3}r\sin\theta\ \frac{1}{2}r^2d\theta}{\int_{-\alpha/2}^{\alpha/2}\ \frac{1}{2}r^2d\theta}=\frac{\frac{r^3}{3}\int_{-\alpha/2}^{\alpha/2}\sin\theta\ \ d\theta}{\frac{r^2}{2}\int_{-\alpha/2}^{\alpha/2}\ \ d\theta}=\frac{\frac{r^3}{3}[-\cos\theta]_{-\alpha/2}^{\alpha/2}}{\frac{r^2}{2}[\theta]_{-\alpha/2}^{\alpha/2}}=0$$




### Example 4

Locate the centroid of the parabolic spandrel formed by a parabola $y=kx^2$ and the vertical line $x=a$.

$$y=kx^2 \gg when\ x=a, y=b \gg b=ka^2 \gg k=\frac{b}{a^2} \gg y=\frac{b}{a^2}x^2$$




Consider an infinitesimal element of thickness $dx$ and height $y$ at a distance $x$ from the origin. Let the centre of this rectangular element be at $(x_i, y_i)$

$$x_i\cong x \quad \ll\gg \quad y_i=\frac{y}{2}=\frac{\frac{b}{a^2}x^2}{2}=\frac{b}{2a^2}x^2 \quad \ll\gg \quad dA=y\,dx=\frac{b}{a^2}x^2\,dx$$

The centroid is given by

$$x_c=\frac{\int x_i\,dA}{\int dA}=\frac{\int_0^a x\ \frac{b}{a^2}x^2\,dx}{\int_0^a\ \frac{b}{a^2}x^2\,dx}=\frac{\frac{b}{a^2}\left[\frac{x^4}{4}\right]_0^a}{\frac{b}{a^2}\left[\frac{x^3}{3}\right]_0^a}=\frac{\frac{a^2b}{4}}{\frac{ab}{3}}=\frac{3a}{4}$$

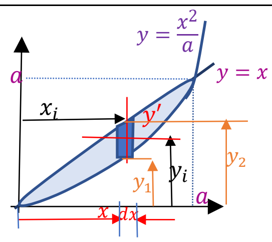
(The integration is with respect to $x$ which varies from $0$ to $a$)

$$y_c=\frac{\int y_i\,dA}{\int dA}=\frac{\int_0^a\left(\frac{b}{2a^2}x^2\right)\frac{b}{a^2}x^2\,dx}{\int_0^a\ \frac{b}{a^2}x^2\,dx}=\frac{\frac{b^2}{2a^4}\left[\frac{x^5}{5}\right]_0^a}{\frac{b}{a^2}\left[\frac{x^3}{3}\right]_0^a}=\frac{\frac{ab^2}{10}}{\frac{ab}{3}}=\frac{3b}{10}$$

### Example 5

Locate the centroid of the shaded area formed by a parabola $y=kx^2$ and the line $y=x$.

$$y=kx^2 \gg when\ x=a, y=a \gg a=ka^2 \gg k=\frac{a}{a^2} \gg y=\frac{x^2}{a}$$




Consider an infinitesimal element of thickness $dx$ and height $y'$ at a distance $x$ from the origin. Let the centre of this rectangular element be at $(x_i, y_i)$.

$$y'=y_2-y_1=x-\frac{x^2}{a}$$

$$x_i\cong x \quad \ll\gg \quad y_i=y_1+\frac{y'}{2}=\frac{x^2}{a}+\frac{x-\frac{x^2}{a}}{2}=\frac{x}{2}+\frac{x^2}{2a} \quad \ll\gg \quad dA=y'\,dx=\left(x-\frac{x^2}{a}\right)dx$$

$$x_c=\frac{\int x_i\,dA}{\int dA}=\frac{\int_0^a x\left(x-\frac{x^2}{a}\right)dx}{\int_0^a\left(x-\frac{x^2}{a}\right)dx}=\frac{\frac{a^3}{12}}{\frac{a^2}{6}}=\frac{a}{2}$$

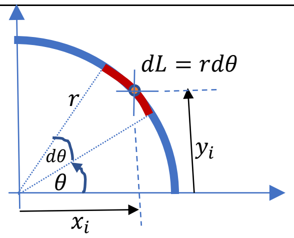
(The integration is with respect to $x$ which varies from $0$ to $a$)

$$y_c=\frac{\int y_i\,dA}{\int dA}=\frac{\int_0^a\left(\frac{x}{2}+\frac{x^2}{2a}\right)\left(x-\frac{x^2}{a}\right)dx}{\int_0^a\left(x-\frac{x^2}{a}\right)dx}=\frac{\frac{1}{2}\left[\frac{x^3}{3}-\frac{x^5}{5a^2}\right]_0^a}{\frac{a^2}{6}}=\frac{2a}{5}$$

(The integration is with respect to $x$ which varies from $0$ to $a$)

## Example 6

Locate the centroid of the quarter circular arc with respect to the co-ordinate axes.




Consider an infinitesimal element of length $dL$ making an included angle $d\theta$ at the origin. Let the centre of this elemental length be at $(x_i, y_i)$.

$$x_i \cong r\cos\theta \quad \lll\ggg \quad y_i \cong r\sin\theta \quad \lll\ggg \quad dL = rd\theta$$

$$x_c = \frac{\int x_i\, dA}{\int dA} = \frac{\int_0^{\frac{\pi}{2}} r\cos\theta \;\; rd\theta}{\int_0^{\frac{\pi}{2}} \;\; rd\theta} = \frac{r^2\int_0^{\frac{\pi}{2}} \cos\theta \;\; d\theta}{r\int_0^{\frac{\pi}{2}} \;\; d\theta} = \frac{r^2[\sin\theta]_0^{\frac{\pi}{2}}}{r[\theta]_0^{\frac{\pi}{2}}} = \frac{2r}{\pi}$$

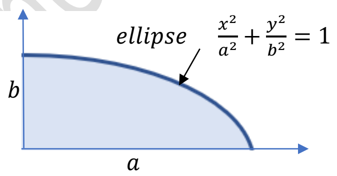
(The integration is with respect to $\theta$ which varies from 0 to $\frac{\pi}{2}$)

$$y_c = \frac{\int y_i\, dA}{\int dA} = \frac{\int_0^{\frac{\pi}{2}} r\sin\theta \;\; rd\theta}{\int_0^{\frac{\pi}{2}} \;\; rd\theta} = \frac{r^2\int_0^{\frac{\pi}{2}} \sin\theta \;\; d\theta}{r\int_0^{\frac{\pi}{2}} \;\; d\theta} = \frac{r^2[-\cos\theta]_0^{\frac{\pi}{2}}}{r[\theta]_0^{\frac{\pi}{2}}} = \frac{2r}{\pi}$$

## Exercise 10

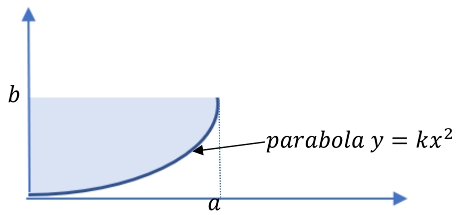
### 10.1

Locate the centroid of the quarter elliptical area with respect to the co-ordinate axes.



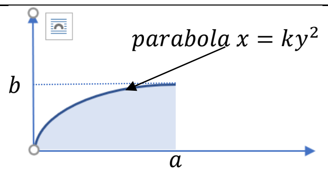

### 10.2

Locate the centroid of the shaded area with respect to the co-ordinate axes.



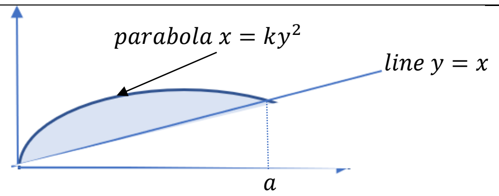

### 10.3

Locate the centroid of the shaded area with respect to the co-ordinate axes.




### 10.4

Locate the centroid of the shaded area with respect to the co-ordinate axes.




Reference

1. Engineering Mechanics: Timoshenko and Young, McGraw Hill
2. Mechanics for Engineers: Beer & Johnston, McGraw Hill
3. Engineering Mechanics: Merriam & Kraig, Wiley
4. Engineering Mechanics: Hibbeler, Pearson

Reference

1. Engineering Mechanics: Timoshenko and Young, McGraw Hill
2. Mechanics for Engineers: Beer & Johnston, McGraw Hill
3. Engineering Mechanics: Merriam & Kraig, Wiley
4. Engineering Mechanics: Hibbeler, Pearson